Regional Ingress Headroom Review

July 2026 operating view | Owner: Samir Holt, Edge SRE | Next review: 2026-07-27

Decision at a glance

Decision
Purchase an additional 10 Gb/s West Hub transit commit while retaining Central Hub as diversion reserve. The cost is justified by regional asymmetry, not total fleet utilization; tunnel loss and convergence remain separate operational gates.

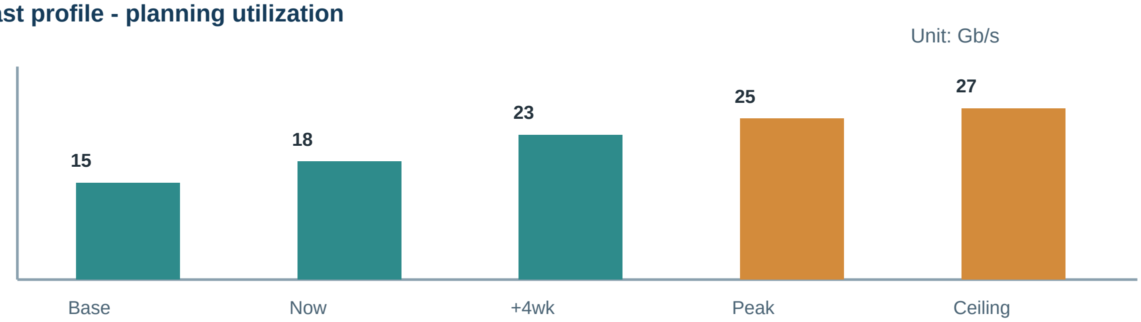
Planning peak
26.7 Gb/s

Evidence
Moderate; peak blend includes weather-diversion scenario

Scope and assumptions

Ingress headroom is calculated across Central Hub, West Hub, and the synthetic rail-status feed. The scenario assumes one region absorbs 35% of a neighbor diversion and encrypted tunnel overhead stays at 6%. It does not model a provider-wide loss because that exceeds the local transit contract decision.

Forecast profile - planning utilization



Capacity signals

Measure	Current	Planning limit	Decision signal
West Hub commit	18.4 Gb/s	26.7 Gb/s	Order new transit when 95th percentile exceeds 22 Gb/s.
Central diversion room	38%	>= 30%	Do not borrow Central reserve for planned maintenance.
Tunnel loss	0.08%	< 0.20%	Loss indicates path quality before bandwidth exhaustion.
Route convergence	14 s	< 30 s	Slow convergence requires routing review, not capacity spend.

Plan, controls, and ownership

Action plan

Action	Due	Owner	Acceptance evidence
Transit amendment	Jul 18	Edge procurement	Add a 10 Gb/s West Hub commitment.
Diversion drill	Jul 21	Network SRE	Shift one-third of synthetic edge load to Central.
Path baseline	Jul 24	Carrier manager	Capture tunnel loss and jitter by ingress class.
Headroom signoff	Jul 27	Samir Holt	Confirm 30% Central reserve after amendment.

Risk register

Risk	Likelihood	Impact	Response
Weather diversion	Medium	High	Preserve Central routing reserve during storms.
Carrier congestion	Medium	Medium	Escalate on loss rather than raw packet rate.
Route flap	Low	High	Validate convergence timers before a transit cutover.

Decision rationale

Purchase an additional 10 Gb/s West Hub transit commit while retaining Central Hub as diversion reserve. The cost is justified by regional asymmetry, not total fleet utilization; tunnel loss and convergence remain separate operational gates.

Selective cross-references

Review alongside: Holiday readiness capacity review and Checkout demand forecast. These linked plans are relevant because they share a limiting dependency or coordinated operating window.

Model notes: all values are synthetic planning figures. No customer records, production identifiers, secrets, or routable network addresses are included. Northstar Transit Cloud | Synthetic regional traffic model